

l.a.

Input a	Input b	Input c	Input d	Output f
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

b.

Input a	Input b	Input c	Output f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

2. Canonical sum-of-products = $x'y'z' + x'y'z + x'yz + xy'z' + xyz = F(x,y,z)$
 Canonical product-of-sum = $(x+y+z) \cdot (x'+y+z') \cdot (x'+y'+z') = F(x,y,z)$

3. Simplify: $x'y'z' + x'y'z + x'yz + xy'z' + xyz$
 $= x'y'(z+z') + yz(x+x') + xy'z' \quad (\text{distributive})$
 $= x'y' + yz + xy'z' \quad (\text{inverse})$

4. Canonical sum-of-products =
 $w'x'y'z' + w'x'y'z + w'x'yz + w'xy'z' + w'xyz + wx'y'z' + wxy'z' + wxyz = F(w,x,y,z)$
 Canonical product-of-sums =
 $(w+x+y+z) \cdot (w+x'+y+z') \cdot (w+x'+y'+z) \cdot (w'+x+y+z') \cdot (w'+x+y'+z) \cdot (w'+x'+y'+z) = F(w,x,y,z)$

5. Simplification
 $F(w,x,y,z) = w'x'y'z' + w'x'y'z + w'x'yz + w'xyz + wx'y'z' + wxy'z' + wxy'z + wxyz$
 $= w'x'y'z' + w'x'y'z + w'x'yz + w'xyz + wx'y'z' + wxy'z' + wxy'z + wxyz$ [Associative]
 $= w'x'y'(z'+z) + w'y'z(x'+x) + wy'z'(x'+x) + wxy'z + w'xyz$ [Inverse]
 $F(w,x,y,z) = w'x'y'(1) + w'y'z(1) + wy'z'(1) + wxy'z + w'xyz$ [Identity]

6.a.

a	b	c	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$\begin{aligned}
 F(a,b,c) &= ab(c+c') + a'c(b+b') \quad (\text{CSOP}) \\
 &= abc + abc' + a'bc + a'b'c \\
 &= \Sigma m(7,6,3,1)
 \end{aligned}$$

$$\begin{aligned}
 F(a,b,c) &= (a+b+c) \cdot (a+b'+c) \cdot (a'+b+c) \cdot (a'+b+c') \\
 &= \Pi M(0,2,4,5)
 \end{aligned}$$

6. b.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

$$\begin{aligned}
 \text{CSOP} &= a'bc'd + abcd' + ab'cd' + ab'cd + abc'd + \\
 &\quad abc'd' + abcd'
 \end{aligned}$$

$$\begin{aligned}
 \text{CPOS} &= (a+b+c+d) \cdot (a+b+c'+d) \cdot (a+b+c'+d') \cdot (a+b'+c+d) \\
 &\quad \cdot (a+b'+c'+d) \cdot (a'+b+c+d) \cdot (a'+b+c+d') \cdot \\
 &\quad (a'+b'+c'+d')
 \end{aligned}$$

$$\begin{aligned}
 7a. \quad F(x, y, z) &= xy'z' + xy'z + x'y'z + x'y'z' + xyz \\
 &= xy'z' + xy'z + x'y'z + xyz + x'y'z' \quad [\text{Associative}] \\
 &= xz'(y+y') + yz(x'+x) + x'y'z' \\
 &= xz'(1) + yz(1) + x'y'z' \quad [\text{Inverse}] \\
 F(x, y, z) &= xz' + yz + x'y'z' \quad [\text{Identity}]
 \end{aligned}$$

$$\begin{aligned}
 b. \quad F(w, x, y, z) &= wxy'z + wxyz' + wxy'z' + wx'y'z + wx'y'z' + wx'y'z' + w'xy'z + w'x'y'z + w'x'y'z' \\
 &= wxy'z + wxyz' + wxy'z' + wx'y'z' + w'x'y'z + w'x'y'z' + wxyz' + w'x'y'z + w'x'y'z' \quad [\text{Associative}] \\
 &= wxy'(z+z') + wx'y'(z+z') + w'x'y'(z+z') + wxyz' + w'x'y'z + w'x'y'z' \\
 &= wxy'(1) + wx'y'(1) + w'x'y'(1) + wxyz' + w'x'y'z + w'x'y'z' \quad [\text{Inverse}] \\
 &= wxy' + wx'y' + w'x'y' + wxyz' + w'x'y'z + w'x'y'z' \quad [\text{Identity}] \\
 &= wy'(x+x') + w'x'y' + wxyz' + w'x'y'z + w'x'y'z' \\
 &= wy'(1) + w'x'y' + wxyz' + w'x'y'z + w'x'y'z' \quad [\text{Inverse}] \\
 F(w, x, y, z) &= wy' + w'x'y' + wxyz' + w'x'y'z + w'x'y'z' \quad [\text{Identity}]
 \end{aligned}$$

$$\begin{aligned}
 c. \quad F(w, x, y, z) &= w'x'y' + w'xz + wxz + w'x'y'z' \\
 &= w'x'y' + xz(w'+w) + w'x'y'z' \\
 &= w'x'y' + xz(1) + w'x'y'z' \quad [\text{Inverse}] \\
 F(w, x, y, z) &= w'x'y' + xz + w'x'y'z' \quad [\text{Identity}]
 \end{aligned}$$

$$d. (x+y+z+w')(v+x)(v'+y+z+w')$$

$$= (x + (y+z+w'))(v)(v'+y+z+w')$$

$$= (x + yv + zv + w'v)(v' + y + z + w')$$

$$= xv' + xy + xz + xw' + yvv' + yvy + yvz + yvw' +$$

$$zv' + zvy + zvz + zvw' + w'vv' + w'vy + w'vz + w'vw'$$

$$= xv' + xy + xz + xw' + yv + yvz + yvw' + zvy + zv + zvw'$$

$$+ w'vy + w'vz + w'$$

$$= xv' + xy + xz + xw' + yv + vz(y+y+1) + w'v(y+y+z+z+1)$$

$$= xv' + xy + xz + xw' + yv + vz + w'v.$$